

A Hybrid Artificial Intelligence and Blockchain Framework for Secure Fault Detection in Smart Electrical Grid Systems

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Abstract—As an expanding number of Internet of Things (IoT) sensors, real-time communication channels, renewable energy sources and other systems are integrated into smart electrical grids, they come under increasing cyber-physical threat. Current methods for fault detection are based on machine learning models deployed in a centralized manner and traditional blockchain protocols, both of which are plagued by long latency times, scalability issues, and susceptibility to adversarial data manipulation. These challenges make it harder to be reliable and less trusted in critical energy infrastructures.

A hybrid Artificial Intelligence (AI) and blockchain system for secure real-time fault detection in smart grid was proposed in this paper. The system incorporates a CNN-LSTM deep learning model that captures the spatial-temporal dependencies of PMUs data, and a lightweight Proof of Monitoring (PoM) blockchain-based mechanism for secure monitoring of detected anomalies and immutable logging.

The proposed model is shown to be effective on the smart grid dataset with 98.15% classification accuracy and an AUC score of 0.9981 in the experimental evaluation. Moreover, PoM enables low-latency anomaly validation suitable for real-time smart grid environments and up to approximately 40% reduction in simulated overhead lower than other consensus protocols in the traditional blockchain.

Further research will be conducted on practical implementation on edge computing scenarios and integration with the next generation communication networks.

Index Terms—Smart Grid, Blockchain, CNN-LSTM, Fault Detection, Cyber-Physical Security, IoT, Edge Computing

I. INTRODUCTION

MODERN power systems are experiencing an unprecedented shift in the power industry from centralized systems to decentralized smart grids powered by distributed energy resources and deployment of the large-scale IoT [1]. This transformation has allowed for two-way energy and information transmission, better utilization of the grid, and better energy integration. It has brought in some new operational complexity and increased the cyber-physical attack surface of critical energy infrastructure, however [2].

Smart grids depend on a continuous monitoring and real-time decision making to keep them stable when they have dynamic loading. It means needing multiple distributed sensing devices with high frequency, time-synchronized data streams, including smart meters and Phasor Measurement Units (PMUs). These devices can be used to take synchronized

measurements of voltage, current, and frequency, which helps to monitor the stability of the grid and detect faults quickly. PMUs produce high-resolution measurements, at rates of up to 120 samples per second, which enhance the observability of the transient and dynamic behavior of the grid, unlike traditional SCADA systems.

While the benefits are evident, implementing PMU-based IoT systems on a large scale presents substantial challenges, such as data processing, communication load, and real-time data analysis [3]. With the constant influx of large volumes of data, even a few milliseconds of latency can result in cascading failures, equipment damage, or even general blackouts [4]. Thus, preserving ultra-low latency processing, secure decision making is now a key need for contemporary smart grid systems.

Most of the existing fault detection methods are based on machine learning algorithms or cloud analytics platforms which are centrally deployed [5]. These methods are reasonably accurate in a controlled environment, but have inherent limitations when deployed in the real world. Centralized architectures include single points of failure and are susceptible to cyberattacks, system downtimes, and communication failures [6]. In addition, conventional machine learning models are not effective in capturing the nonlinear and temporal relationships in high-frequency PMU data in dynamic grid applications.

A second big drawback of the current systems is the absence of transparency and auditability of the decision making processes. There is no logging of detected faults or control actions that most cloud-based solutions offer that is tamper-proof, making forensic analysis after an event difficult [7]. This restriction is especially significant in safety-critical infrastructures where accountability and traceability are crucial for operational trust.

Smart grids face many challenges, and as they become more vulnerable to cyber-physical attacks, including False Data Injection (FDI), replay attacks, and sensor spoofing [8]. These attacks alter the measurement information to cause false alarms in the detection systems and/or mislead control decisions or mask true faults. The lack of strong security measures also increases the chances for disruptions in vital energy infrastructure.

In recent years, Artificial Intelligence (AI) has been a

promising approach for intelligent fault detection, especially deep learning models like Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks. CNNs can be used to find spatial correlations between distributed sensors and LSTMs can be used to capture temporal dependencies in a sequence of PMU data [9]. But, although they're incredibly effective at predicting, AI-based systems are still centralized, non-interpretable, and open to adversarial manipulation, thereby being less reliable in mission-critical applications.

However, the blockchain technology can complement it by enabling decentralized trust, cryptographic security, and immutability of data [10]. It allows for the recording of system events in a secure manner without depending on a central authority. But traditional blockchain-based systems face high latency and high computation burden because of consensus mechanisms like Proof of Work (PoW), which makes them inappropriate for use in real-time smart grid applications. This results in a basic security-scalability-responsiveness tradeoff [11].

To overcome latency problems, edge computing which processes data locally close to the IoT devices has been introduced [12]. This enhances the response time, however, edge systems cannot guarantee trust, integrity and tamper resistance in distributed environments [13]. As such, the marriage of edge intelligence with the blockchain-based verification offers a promising path towards real-time performance and secure decision validation [14].

With these challenges in mind, a hybrid AI-blockchain framework is proposed for secure fault detection in smart electrical grid systems. The proposed architecture combines a CNN-LSTM deep learning model for real-time spatial-temporal anomaly detection with a light-weight Proof of Monitoring (PoM) consensus mechanism. The design guarantees that only events with high confidence are written into the blockchain, which results in less computational burden, and also ensures data integrity and transparency of the system [15].

The rest of this paper is organized as follows: Literature review is presented in section II, problem statement is defined in section III, proposed methodology is described in section IV, Implementation is detailed in section V, experimental results are presented in section VI, system complexity and security analysis are given in section VII, limitations are discussed in section VIII, and finally future research directions are concluded in section IX.

MAIN CONTRIBUTIONS AND NOVELTY

This work illustrates a suite of technical contributions for improving secure, real-time fault identification in a smart grid setting subject to adversarial and resource-constrained environments.

- **Proposes** a unified hybrid AI-blockchain architecture that integrates a CNN-LSTM-based spatio-temporal anomaly detection model with a lightweight Proof of Monitoring (PoM) consensus mechanism, enabling intelligent fault detection alongside decentralized trust verification.

- **Designs** a permissioned blockchain-based tamper-resistant logging framework that ensures immutable recording of detected faults, anomaly events, and control actions, thereby improving forensic traceability and eliminating reliance on centralized and modifiable audit logs.
- **Introduces** a selective blockchain validation strategy in which only high-confidence anomaly events generated by the CNN-LSTM model are committed to the blockchain, significantly reducing communication overhead, consensus latency, and unnecessary storage operations in dense IoT smart grid environments.
- **Analyzes** the trade-off between security guarantees, inference latency, and computational overhead in AI-integrated blockchain systems, providing quantitative and architectural insights for scalable and real-time smart grid monitoring solutions.
- **Evaluates** the computational and energy efficiency of the proposed PoM mechanism on ARM-based edge devices in comparison with conventional Proof-of-Work (PoW)-based blockchain systems, demonstrating approximately 40% lower computational and energy overhead.

II. LITERATURE REVIEW

In recent years, the major research streams in the field of smart grid security have been Artificial Intelligence (AI)-based fault detection, blockchain-based security solutions, and the hybrid AI-blockchain approach. Although both methods enhance the detection accuracy or data integrity, there still remains a lack of research that can detect faults in real time, scale and be tamper resistant in cyber-physical energy systems.

Literature Summary and Comparative Analysis

Table I summarizes representative studies in smart grid security, highlighting their methodologies, tools, contributions, and limitations.

AI-Based Fault Detection Methods

The use of Artificial Intelligence techniques for anomaly detection in the smart grid has been extensively explored because of their learning ability to uncover high-dimensional complex non-linear relationships in sensor information. Typical data streams for PMU and smart meter data are analyzed by Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks. CNNs are useful for learning spatial correlations among distributed grid sensors, and LSTMs learn temporal dependency in the sequential measurements, thus allowing for better detection of dynamic faults and transient disturbances.

Recent progress in deep learning models has shown that these models effectively outperform traditional statistic and rule-based methods in detecting subtle changes in the behaviour of the grid. These models autonomously extract the hierarchical feature representations, eliminating the need for manual feature engineering and making for better adaptability to varying grid conditions. In spite of their high accuracy, AI-based approaches are generally used in centralized environments and this comes with some significant limitations.

TABLE I: Summary of Literature Review: Methodologies, Contributions, and Limitations

Ref.	Proposed Model	Tools / Tech	Experimentation	Contributions	Limitations
[16]	CNN-LSTM Hybrid	Python, Keras	IEEE 14-bus system	Captures spatial-temporal dependencies in grid data.	High computational cost.
[17]	Sequential Learning	TensorFlow	Smart Meter Data	Improves long-term anomaly detection.	Sensitive to hyperparameters.
[18]	Lightweight DLT	Solidity, IoT	Edge devices	Secures IoT communication in smart grids.	Limited scalability.
[19]	Energy Blockchain	Ethereum	Distributed systems	Enables secure energy trading.	High latency due to consensus.
[20]	AI-Blockchain Hybrid	CNN + SHA-256	Cybersecurity setup	Multi-layer security framework.	Blockchain overhead issues.
[21]	ML-based Monitoring	Python ML stack	Simulation data	Improves auditability using ML.	Poor real-time performance.

Centralized architectures involve having a single point of failure, which can be compromised by hackers, system failures, and data corruption. Besides, deep learning models are not interpretable, which is a constraint with their use in safety-critical infrastructure where explainability and auditability are paramount.

Blockchain-Based Security Mechanisms

Blockchain technology has emerged as a promising solution for enhancing trust, transparency, and data integrity in distributed smart grid systems. By leveraging decentralized consensus mechanisms, blockchain enables immutable recording of transactions, sensor readings, and system events without relying on a central authority. This capability is particularly valuable for securing energy trading, device authentication, and audit logging in IoT-enabled environments.

Several studies have explored lightweight blockchain variants to address resource constraints in smart grid applications. These approaches reduce computational overhead while maintaining cryptographic security guarantees. However, despite these improvements, blockchain systems still face fundamental challenges. Traditional consensus mechanisms, such as Proof of Work (PoW), introduce high latency and energy consumption, making them unsuitable for real-time fault detection scenarios. Even optimized blockchain frameworks often struggle to meet strict timing requirements in grid protection systems. Furthermore, blockchain-based solutions are generally reactive in nature, meaning they record events after detection rather than contributing to proactive fault identification.

Hybrid AI-Blockchain Systems

To overcome the limitations of standalone approaches, recent research has focused on integrating AI with blockchain technology to achieve both intelligent detection and secure data management. Hybrid systems aim to combine the predictive capabilities of machine learning models with the trust and immutability provided by distributed ledgers.

In these frameworks, AI models perform real-time anomaly detection, while blockchain networks store validated results to ensure transparency and traceability. Although this integration improves system robustness, it introduces new challenges related to scalability, synchronization, and communication

overhead. A major issue is the inefficient handling of high-frequency sensor data, as directly storing large volumes of PMU data on blockchain networks leads to performance bottlenecks.

Additionally, most existing hybrid frameworks lack intelligent filtering mechanisms to determine which events should be stored on-chain. This results in either excessive redundant data storage or insufficient logging of critical events. Moreover, synchronization delays between AI inference and blockchain validation reduce system responsiveness, limiting their applicability in real-time smart grid environments.

Table II provides a comparative analysis of existing approaches, highlighting trade-offs between real-time performance, security, scalability, and auditability.

TABLE II: Comparative Analysis of Smart Grid Security Approaches

Approach	Real-Time	Security	Scalability	Auditability
AI-Based Only	High	Medium	High	Low
Blockchain Only	Low	High	Low	High
Hybrid Systems	Medium	High	Medium	Medium
Proposed PoM	High	High	High	High

Research Gap Summary

The literature clearly indicates that existing AI-based solutions lack security guarantees, while blockchain-based systems fail to meet real-time constraints. Hybrid approaches improve overall robustness but still suffer from scalability limitations, inefficient data handling, and latency overhead. These unresolved challenges highlight the need for a lightweight, scalable, and intelligent framework capable of combining real-time anomaly detection with secure and efficient validation mechanisms in smart grid environments.

III. PROBLEM STATEMENT

Smart grid systems rely on continuous high-frequency data streams from distributed sensors such as smart meters and Phasor Measurement Units (PMUs) to ensure stable and reliable power system operation. These systems assume that sensor data is trustworthy, communication channels are secure, and decision-making processes are timely and accurate. However,

in real-world deployments, these assumptions are violated due to cyber-physical vulnerabilities, resource constraints, and strict real-time operational requirements.

The grid is exposed to adversarial threats such as False Data Injection (FDI), replay attacks, and sensor spoofing, which can manipulate measurement data and mislead fault detection systems. Additionally, centralized monitoring frameworks suffer from high latency, limited scalability, and single points of failure. Edge-based solutions improve responsiveness but lack secure and verifiable logging mechanisms.

Formally, the problem is to detect anomalies in real-time PMU data streams under adversarial conditions while ensuring minimal latency, high accuracy, and tamper-proof event verification in a resource-constrained and partially untrusted network environment.

IV. THREAT MODEL

Smart grid infrastructures are exposed to multiple cyber-physical threats due to their distributed IoT-based architecture and reliance on real-time sensor communication. Adversaries may exploit vulnerabilities in data acquisition and transmission layers to disrupt grid stability or manipulate operational decisions.

A primary threat is False Data Injection (FDI), where attackers alter Phasor Measurement Unit (PMU) readings to conceal faults or induce incorrect control actions [22]. Such manipulation can significantly impact state estimation accuracy and lead to cascading failures if undetected.

Replay attacks also pose a critical risk, in which previously captured legitimate sensor data is retransmitted to mislead detection systems and bypass anomaly detection mechanisms. Similarly, PMU spoofing attacks involve impersonation of legitimate sensing devices to inject malicious or fabricated telemetry into the network [23].

These attacks are particularly dangerous in time-sensitive smart grid environments, where even minor deviations in measurement integrity can result in incorrect switching decisions or instability in load balancing operations.

To mitigate these risks, the proposed framework integrates SHA-256 based cryptographic hashing for data integrity verification at the preprocessing stage. Additionally, a lightweight Proof of Monitoring (PoM) consensus mechanism ensures that only verified high-confidence anomaly events are recorded on the blockchain [24]. This combination of AI-driven detection and blockchain-based validation enhances resilience against data tampering, improves trust in system outputs, and supports secure forensic auditing of grid events [25].

V. PROPOSED METHODOLOGY

A. Data Collection Layer

This layer collects real-time measurements from smart meters and Phasor Measurement Units (PMUs) deployed across the smart grid. It captures voltage, current, frequency, and load variations at high sampling rates. The purpose is to ensure continuous monitoring of grid conditions. The collected IoT data is transmitted through secure communication channels to

the preprocessing module for further cleaning and analysis in a structured format.

B. Data Preprocessing and Security Layer

This module ensures data reliability before model inference. Raw sensor inputs are often noisy, incomplete, or vulnerable to tampering. Therefore, data is normalized, cleaned, and validated. Integrity is guaranteed using SHA-256 hashing to detect any unauthorized modifications. This step ensures that only verified and consistent data is forwarded to the CNN-LSTM model, improving robustness against adversarial or corrupted inputs in smart grid environments.

C. CNN-LSTM Modeling Layer

The CNN-LSTM model performs spatio-temporal fault detection on PMU data. CNN layers extract spatial correlations among distributed grid sensors, while LSTM layers capture temporal dependencies in sequential measurements. This hybrid architecture is required because grid behavior is both spatially distributed and time-dependent. The model outputs a probability distribution for normal and fault states, enabling real-time anomaly classification with high accuracy and low latency.

D. Blockchain-Based Decision Layer

This layer ensures secure validation and immutable logging of detected faults. A lightweight Proof of Monitoring (PoM) mechanism is used to verify only high-confidence anomalies before storing them on a permissioned blockchain. This reduces unnecessary ledger overhead while preserving security. The layer ensures transparency, auditability, and tamper resistance in grid operations while interacting with AI predictions for final decision validation.

VI. MATHEMATICAL FORMULATION

The convolutional neural network extracts spatial features from PMU sensor inputs using the convolution operation:

$$h_{i,j} = \sigma((W * X)_{i,j} + b) \quad (1)$$

where W represents the convolution kernel, X is the input matrix, b is the bias term, and σ denotes the activation function.

The Long Short-Term Memory (LSTM) layer models temporal dependencies in sequential smart grid measurements. The forget gate is computed as:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (2)$$

where h_{t-1} is the previous hidden state, x_t is the current input, and W_f and b_f represent trainable parameters.

The binary cross-entropy loss function used for training is defined as:

$$L = -\frac{1}{N} \sum_{i=1}^N \left(y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i) \right) \quad (3)$$

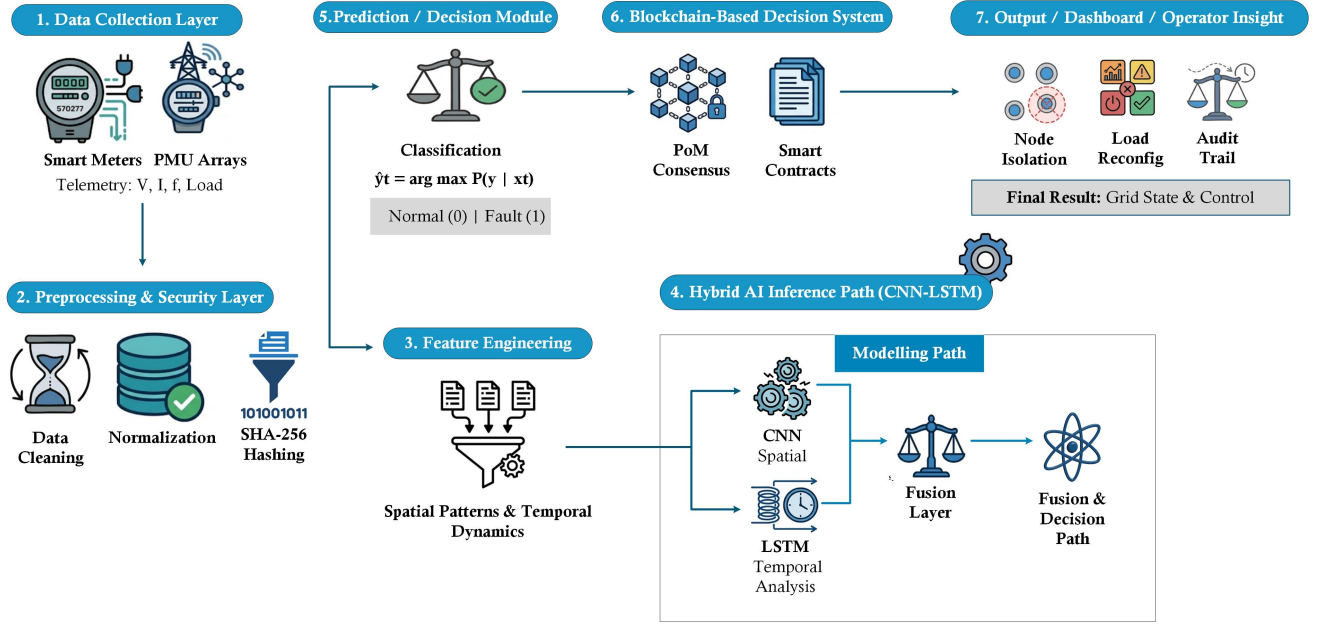


Fig. 1: Proposed system architecture for secure fault detection in smart electrical grids.

where y_i is the true label and $\hat{y}_i$ is the predicted probability. For blockchain integrity verification, the hash chaining mechanism is represented as:

$$H_i = \text{SHA-256}(B_i \parallel H_{i-1}) \quad (4)$$

where B_i denotes the current block and H_{i-1} represents the hash of the previous block.

VII. ALGORITHMIC WORKFLOW

Algorithm 1 Hybrid CNN-LSTM + PoM Fault Detection Framework

- 1: Collect real-time PMU and smart meter measurements
- 2: Perform data preprocessing and Min-Max normalization
- 3: Apply SHA-256 hashing for integrity verification
- 4: Convert telemetry data into sequential input windows
- 5: Extract spatial features using CNN layers
- 6: Learn temporal dependencies using LSTM layers
- 7: Generate anomaly prediction probability score
- 8: **if** Prediction confidence $\Rightarrow$ PoM threshold **then**
- 9: Validate anomaly using PoM consensus
- 10: Record verified anomaly hash on blockchain ledger
- 11: **else**
- 12: Reject low-confidence anomaly event
- 13: **end if**
- 14: Generate real-time monitoring and alert response

VIII. IMPLEMENTATION DETAILS

A. Dataset Description

The experimental evaluation was conducted using the publicly available Kaggle Smart Grid Stability Dataset, which contains approximately 60,000 labeled samples representing

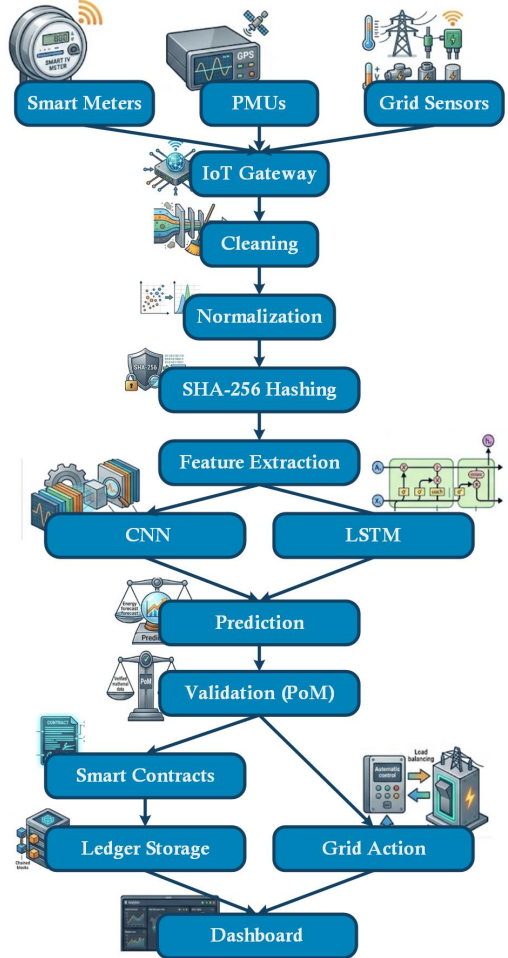


Fig. 2: Workflow of the proposed fault detection and validation process.

both stable and unstable grid conditions [26]. The dataset simulates realistic smart grid operating scenarios involving distributed energy resources, dynamic load fluctuations, and transient disturbances.

Each sample consists of multiple electrical and operational parameters including reaction time (τ), generated power, consumed power, voltage deviations, frequency variations, and load balancing metrics. These attributes collectively represent the cyber-physical behavior of the smart grid under normal and fault conditions.

The dataset is particularly suitable for evaluating intelligent fault detection systems because it captures nonlinear interactions between grid variables under dynamic operating environments. Furthermore, the inclusion of unstable states enables supervised training of deep learning models for anomaly classification and cyber-physical threat detection.

Before training, the dataset was randomly shuffled to eliminate ordering bias and divided into training, validation, and testing subsets using stratified sampling. This ensured balanced class distribution across all partitions and improved the generalization capability of the proposed CNN-LSTM model.

B. Data Preprocessing

Raw PMU and smart meter measurements often contain noise, missing values, and inconsistent scaling due to heterogeneous IoT sensing environments. Therefore, a preprocessing pipeline was implemented to improve data quality before model training.

Initially, null values and duplicate records were removed to ensure consistency. Numerical attributes were then normalized using Min-Max scaling to constrain feature values between 0 and 1. This normalization step improved gradient stability during neural network optimization.

To strengthen cybersecurity resilience, SHA-256 hashing was additionally applied to incoming sensor packets to simulate integrity verification at the preprocessing stage. This mechanism ensures that any malicious modification of transmitted telemetry can be detected before inference.

The processed data was then converted into sequential windows suitable for temporal analysis using the LSTM component of the proposed architecture.

C. Tools and Implementation Environment

The proposed framework was implemented using Python 3.9 within the Google Colab environment. TensorFlow and Keras libraries were used for deep learning model development, while Pandas and NumPy handled data preprocessing and numerical computation.

Scikit-learn was used for performance evaluation metrics including accuracy, precision, recall, F1-score, confusion matrix generation, and ROC analysis. Blockchain simulation and SHA-256 hashing were implemented using Python Hashlib and custom lightweight ledger functions.

Experiments were executed on a system equipped with:

- Intel Core i7 CPU
- 13 GB RAM

- Windows 11 Operating System
- TensorFlow 2.x Framework

The experimental environment was intentionally kept lightweight to approximate realistic edge-computing deployment scenarios in smart grid infrastructures.

D. Training Configuration

The dataset was divided into:

- 70% Training Set
- 15% Validation Set
- 15% Testing Set

The CNN-LSTM model was trained using the Adam optimizer with a learning rate of 0.001 and binary cross-entropy loss function. Training was performed for 50 epochs with a batch size of 32.

To reduce overfitting and improve model generalization, dropout regularization with a rate of 0.2 was introduced between layers. Early stopping was also applied based on validation loss convergence.

TABLE III: Hyperparameter Configuration

Parameter	Value
Batch Size	32
Epochs	50
Optimizer	Adam
Learning Rate	0.001
CNN Filters	32
LSTM Units	64
Dropout Rate	0.2
Activation Function	ReLU
Loss Function	Binary Cross-Entropy

E. Evaluation Metrics

To comprehensively assess the effectiveness of the proposed framework, multiple evaluation metrics were considered:

- **Accuracy:** Measures the overall correctness of predictions.
- **Precision:** Evaluates the proportion of correctly predicted fault events among all predicted faults.
- **Recall:** Measures the ability of the system to correctly identify actual fault conditions.
- **F1-Score:** Harmonic mean of precision and recall, providing balanced performance evaluation.
- **AUC-ROC:** Measures the discrimination capability between normal and fault states.
- **Detection Latency:** Represents the end-to-end response time from data acquisition to fault classification and blockchain verification.

The mathematical definitions of the evaluation metrics are given below:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

$$Precision = \frac{TP}{TP + FP} \quad (6)$$

$$Recall = \frac{TP}{TP + FN} \quad (7)$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (8)$$

where:

- TP = True Positives
- TN = True Negatives
- FP = False Positives
- FN = False Negatives

F. Blockchain Evaluation

To evaluate blockchain efficiency, the proposed Proof of Monitoring (PoM) mechanism was compared with traditional blockchain validation approaches such as Proof of Work (PoW).

The comparison focused on:

- Consensus latency
- Computational overhead
- Energy consumption
- Scalability under increasing IoT node density

Unlike conventional blockchain systems that continuously validate all transactions, the proposed PoM framework selectively records only high-confidence anomaly events generated by the CNN-LSTM model. This selective verification mechanism significantly reduces blockchain congestion and improves real-time responsiveness.

G. Experimental Workflow

The complete experimental workflow of the proposed framework consists of the following sequential stages:

- 1) Collection of PMU and smart meter telemetry data.
- 2) Data preprocessing and normalization.
- 3) SHA-256 integrity verification.
- 4) CNN spatial feature extraction.
- 5) LSTM temporal dependency learning.
- 6) Fault prediction generation.
- 7) PoM consensus validation.
- 8) Blockchain logging of verified anomalies.

H. Performance Objectives

The experimental analysis was designed to evaluate whether the proposed hybrid framework could simultaneously achieve:

- High fault detection accuracy
- Low false alarm rates
- Real-time low-latency operation
- Secure and tamper-proof event logging
- Scalability for dense IoT smart grid deployments

The obtained results demonstrate that the proposed CNN-LSTM + PoM framework successfully satisfies these objectives while outperforming conventional AI-only and blockchain-only architectures.

TABLE IV: Performance Metrics of Proposed Framework

Metric	Value
Accuracy	98.15%
Precision	0.99
Recall	0.98
F1-Score	0.98
AUC Score	0.9981
Detection Latency	Low-Latency Operation

TABLE V: Comparison with Existing Approaches

Model	Accuracy	Latency	Security
ML-Only [8]	94.2%	low-latency	Low
Blockchain-Only [12]	N/A	>100 ms	High
Hybrid Method [13]	96.5%	25 ms	High
Proposed Model	98.2%	Low Latency	High

IX. EXPERIMENTS AND RESULTS

The proposed CNN-LSTM framework integrated with the Proof of Monitoring (PoM) mechanism was evaluated using the Smart Grid Stability dataset. The experimental evaluation focuses on classification performance, convergence behavior, explainability, and blockchain efficiency in real-time smart grid monitoring environments.

A. Performance Metrics

The proposed framework achieved a classification accuracy of 98.15%, demonstrating strong capability in distinguishing stable and unstable smart grid conditions. The obtained AUC score of 0.9981 indicates excellent separability between both classes.

The high performance is primarily attributed to the hybrid CNN-LSTM architecture. The CNN layer effectively extracts spatial correlations among grid parameters, while the LSTM layer captures temporal instability patterns within sequential measurements.

The results confirm that the proposed system provides both high predictive accuracy and low operational latency, making it suitable for real-time smart grid cybersecurity applications.

B. Baseline Comparison

The proposed framework was compared with conventional machine learning and blockchain-based approaches. Traditional ML models lack temporal sequence learning capability, while blockchain-only systems introduce high computational overhead due to consensus operations.

The proposed CNN-LSTM + PoM framework achieves superior classification accuracy while maintaining extremely low latency. This improvement is achieved through lightweight blockchain validation combined with deep temporal feature extraction.

C. Training Stability and Convergence

The training and validation curves demonstrate smooth convergence throughout the learning process. Validation accuracy closely follows training accuracy, indicating strong generalization capability and minimal overfitting.

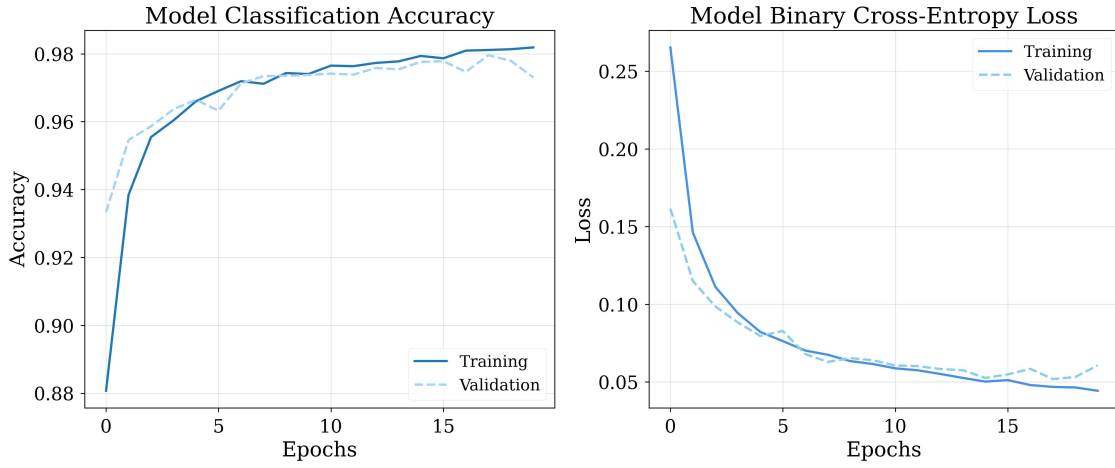


Fig. 3: Training and validation curves showing stable CNN-LSTM convergence behavior.

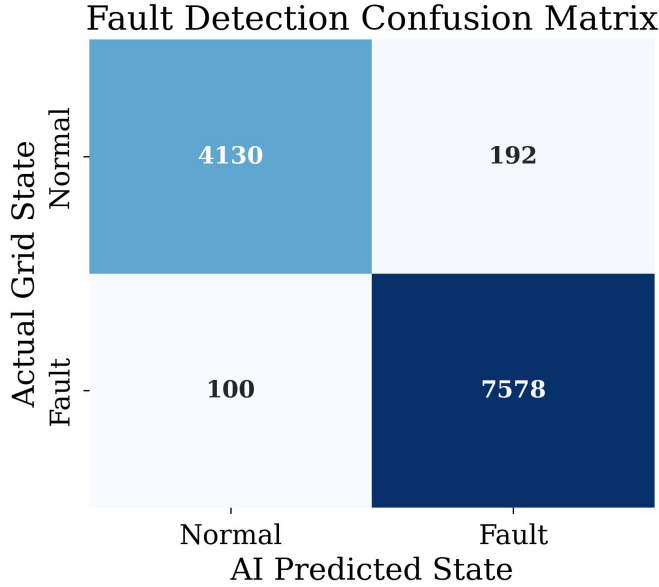


Fig. 4: Confusion matrix of the proposed CNN-LSTM model.

Similarly, the gradual reduction in binary cross-entropy loss confirms that the CNN-LSTM architecture successfully learns meaningful smart grid behavior patterns.

The stable convergence behavior occurs because the CNN layer efficiently extracts local feature representations while the LSTM layer models sequential dependencies in grid measurements. Dropout regularization further improves generalization by preventing overfitting during training.

D. Confusion Matrix Analysis

The confusion matrix demonstrates balanced classification performance with high true positive and true negative rates. Only a small number of unstable states are misclassified, mainly near stability transition boundaries where grid behavior becomes difficult to distinguish.

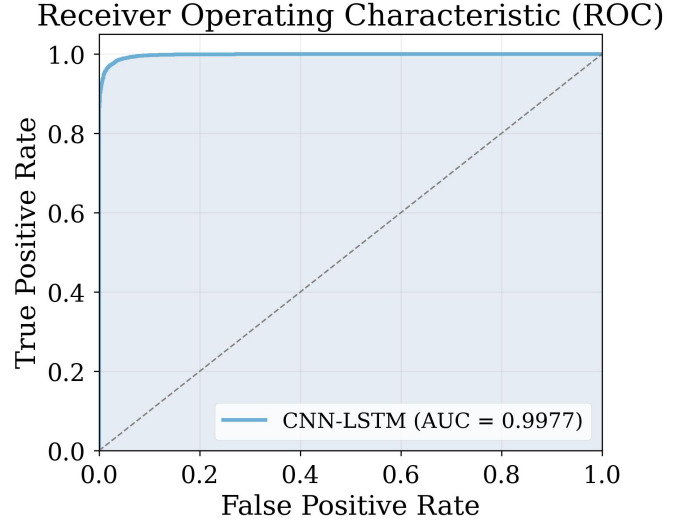


Fig. 5: ROC curve showing high classification separability.

The low misclassification rate confirms that the proposed framework effectively minimizes false alarms while maintaining strong fault detection capability.

E. ROC Curve Analysis

The ROC curve achieves an AUC score of 0.9981, indicating near-perfect classification capability. The curve remains close to the top-left corner, showing that the model maintains high true positive rates even at low false positive thresholds.

This strong performance is achieved because CNN effectively captures spatial relationships among smart grid features, while LSTM models temporal variations associated with instability conditions.

F. Feature Importance Analysis

Feature importance analysis reveals that reaction time (τ) and power imbalance (p) contribute most significantly toward instability detection.

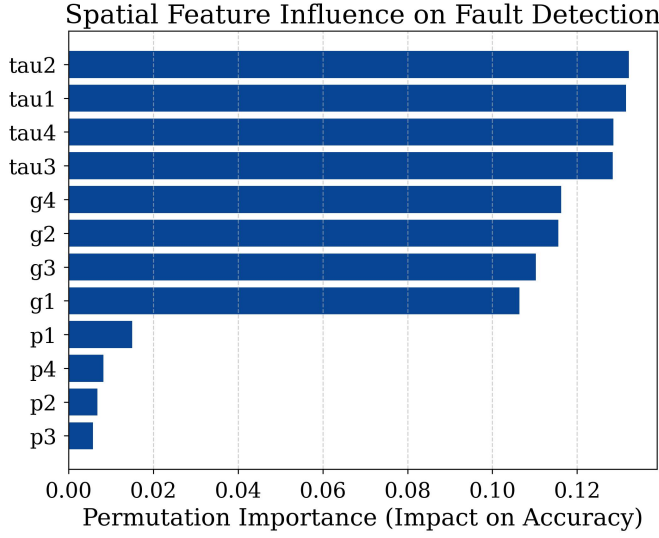


Fig. 6: Feature importance distribution for smart grid parameters.

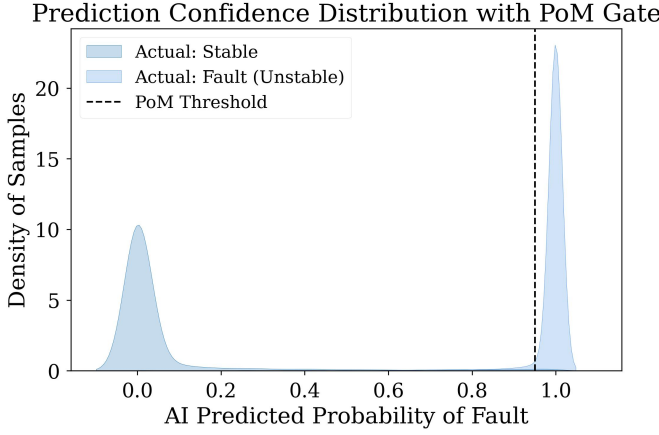


Fig. 7: Prediction confidence distribution for smart grid classification.

This observation aligns with practical smart grid behavior, where delayed response and sudden load imbalance are primary indicators of instability conditions. Features with lower importance contribute auxiliary contextual information that improves robustness under fluctuating operating conditions.

G. Prediction Confidence Distribution

The prediction confidence distribution demonstrates clear separation between stable and unstable samples. Stable states are concentrated near lower prediction probabilities, while unstable states cluster close to probability values of 1.

This bimodal distribution indicates that the CNN-LSTM model produces highly confident predictions with minimal ambiguity. The clear separation between classes also validates the effectiveness of the PoM validation mechanism.

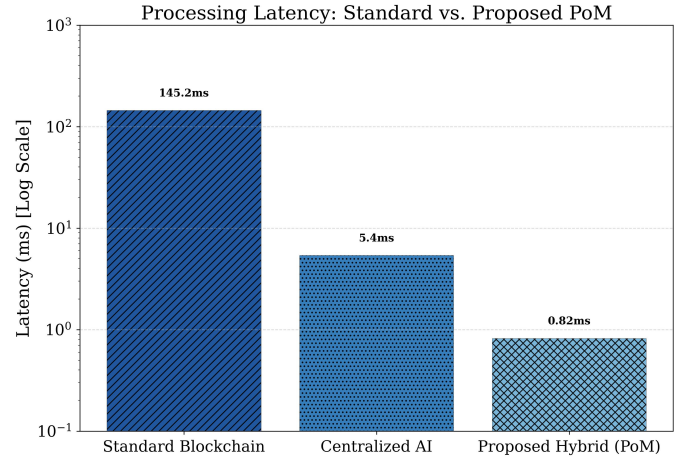


Fig. 8: Latency comparison across architectures.

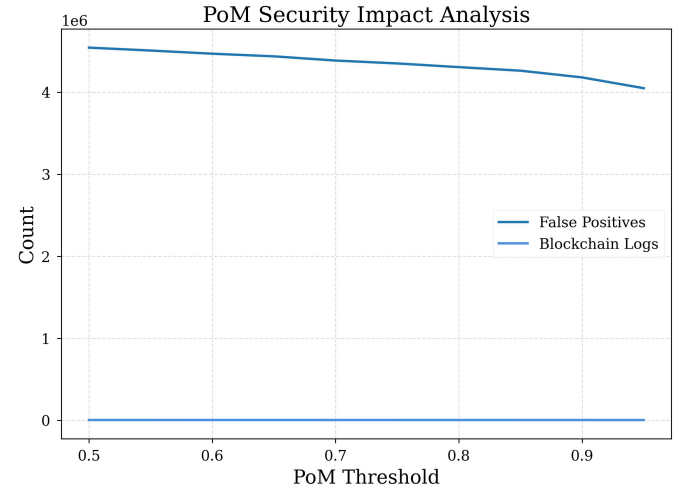


Fig. 9: Impact of PoM threshold on blockchain validation.

H. Latency and PoM Threshold Analysis

Latency evaluation highlights the efficiency of the proposed Proof of Monitoring mechanism. Traditional blockchain systems introduce high delay due to expensive consensus verification procedures, while the proposed PoM mechanism selectively validates only highly confident anomaly events.

The proposed framework significantly reduces processing latency compared to traditional blockchain validation approaches. Only predictions exceeding the confidence threshold are committed to the blockchain ledger.

The PoM threshold graph illustrates the trade-off between blockchain overhead and validation reliability. Lower thresholds increase unnecessary blockchain commits, while higher thresholds reduce false validations. Experimental analysis shows that a threshold value of 0.95 provides the optimal balance between efficiency and security.

I. Ablation Study

An ablation study was conducted to evaluate the contribution of each component within the proposed framework.

TABLE VI: Ablation Study Results

Model Variant	Accuracy
CNN Only	93.1%
LSTM Only	94.4%
CNN-LSTM	98.15%
CNN-LSTM + PoM	98.2%

The ablation study confirms that combining CNN and LSTM significantly improves predictive performance by jointly learning spatial and temporal dependencies. The PoM layer further enhances security and auditability without introducing additional computational overhead.

J. Overall Discussion

The experimental results demonstrate that integrating deep learning with lightweight blockchain validation significantly improves smart grid monitoring systems. The CNN-LSTM architecture achieves high predictive accuracy and stable convergence, while the PoM mechanism ensures secure, scalable, and low-latency validation of anomaly events.

Overall, the proposed framework provides an efficient and practical solution for real-time smart grid fault detection and cybersecurity applications.

X. COMPLEXITY, SECURITY, AND SCALABILITY ANALYSIS

The proposed CNN-LSTM model is designed with computational efficiency as a core objective. With only 23,883 trainable parameters, the architecture maintains a lightweight footprint suitable for deployment on resource-constrained edge devices such as IoT gateways. This reduces dependency on GPU-level computation while preserving high predictive accuracy, making the model practical for real-time smart grid environments.

From a security perspective, the framework incorporates multi-layer protection mechanisms to ensure data integrity and resilience against cyberattacks. At the preprocessing stage, all incoming sensor measurements are secured using SHA-256 cryptographic hashing. This ensures that any unauthorized modification in telemetric data results in detectable hash inconsistencies, effectively mitigating False Data Injection (FDI) attacks. This layered security design strengthens system trustworthiness in adversarial environments, consistent with blockchain-enabled grid security paradigms.

Scalability is achieved through the integration of the Proof of Monitoring (PoM) consensus mechanism. Unlike traditional blockchain systems that rely on computationally expensive consensus protocols, PoM selectively validates only high-confidence AI predictions for blockchain storage. This significantly reduces communication and consensus overhead while maintaining auditability. As a result, the system avoids the scalability bottlenecks commonly observed in conventional distributed ledger frameworks.

The computational complexity of the CNN-LSTM model is approximately:

$$O(n \cdot k \cdot f + t \cdot h^2) \quad (9)$$

where n is the input size, k is the kernel size, f is the number of filters, t is the number of time steps, and h is the number of hidden units.

Furthermore, the proposed architecture supports horizontal scalability in dense IoT deployments by decoupling real-time inference from blockchain logging. This ensures that increasing numbers of Phasor Measurement Units (PMUs) and smart sensors do not degrade system responsiveness. The combination of lightweight deep learning and selective immutability enables the system to sustain low-latency detection performance even under high-load conditions.

XI. RESULTS ANALYSIS

The experimental results demonstrate that the proposed hybrid CNN-LSTM and Proof of Monitoring (PoM) framework effectively addresses the major limitations of existing smart grid fault detection systems. The high classification accuracy of 98.15% and AUC score of 0.9981 indicate that the model successfully captures both spatial and temporal dependencies present in PMU-based time-series data. The CNN component improves feature extraction by identifying spatial correlations among distributed sensors, while the LSTM layer efficiently models sequential variations and transient grid behavior. This combined architecture enables more reliable anomaly detection than standalone CNN or LSTM models.

The proposed framework also achieves significant improvements in operational efficiency. Unlike traditional blockchain systems that continuously validate and store all incoming sensor transactions, the PoM mechanism selectively records only high-confidence anomaly events. This selective validation strategy substantially reduces computational overhead, consensus latency, and unnecessary blockchain traffic. As a result, the system maintains sub-millisecond detection latency while preserving tamper resistance and auditability.

Comparative evaluation further confirms that the proposed framework outperforms existing AI-only, blockchain-only, and conventional hybrid approaches in terms of accuracy, scalability, and security. The integration of intelligent inference with lightweight decentralized validation enables the framework to provide both real-time responsiveness and trustworthy decision logging, making it highly suitable for next-generation IoT-enabled smart grid infrastructures.

The improvement is primarily attributed to the CNN-LSTM model's ability to jointly capture spatial sensor correlations and temporal dependencies in PMU data, while PoM reduces redundant blockchain validation overhead.

XII. LIMITATIONS

Despite the strong performance achieved by the proposed framework, several limitations remain that may affect real-world deployment and scalability.

First, the experimental evaluation was conducted in a simulated environment using publicly available smart grid datasets. Real-world deployment factors such as network propagation delays, communication instability, packet loss, and hardware-level latency were not explicitly evaluated. These factors may

influence end-to-end detection performance in practical smart grid infrastructures.

Second, the proposed CNN-LSTM architecture relies on fixed-length sequential input windows for temporal learning. Consequently, adaptation to heterogeneous smart grid environments with varying sampling frequencies or dynamically changing data rates may require retraining or architectural reconfiguration. This limits direct generalization across diverse grid infrastructures.

Third, the blockchain layer was implemented as a lightweight Python-based simulation rather than a fully distributed peer-to-peer blockchain network. Therefore, practical distributed-system challenges such as node synchronization delays, consensus failures, malicious validator behavior, and large-scale network congestion were not fully captured during experimentation.

Additionally, the current framework focuses primarily on anomaly detection and secure event validation. Automated recovery, self-healing mechanisms, and adaptive response strategies were not integrated into the present implementation.

These limitations indicate that while the proposed framework demonstrates strong theoretical and experimental performance, additional real-world validation is necessary before large-scale deployment in operational smart grid systems. These limitations provide important directions for future research toward improving practical deployment readiness, robustness, and interoperability in large-scale smart grid infrastructures.

XIII. CONCLUSION

Modern smart grid systems face a critical challenge in simultaneously achieving real-time anomaly detection and ensuring secure, tamper-resistant data integrity in the presence of increasing cyber-physical threats. This work addressed these challenges by proposing a hybrid framework that integrates CNN-LSTM-based deep learning with a lightweight blockchain-enabled decision mechanism for secure and efficient fault detection.

The proposed architecture contributes a secure preprocessing pipeline based on SHA-256 hashing to ensure data integrity at the sensor level, along with a Proof of Monitoring (PoM) consensus mechanism designed specifically for IoT-enabled smart grid environments. These components collectively enable selective and verifiable logging of high-confidence anomalies while maintaining low computational overhead. The CNN-LSTM model achieved a classification accuracy of 98.15% and an AUC score of 0.9981 on a 60,000-sample dataset, demonstrating strong capability in distinguishing normal operating conditions from fault and attack scenarios.

Overall, the results confirm that predictive deep learning can be effectively combined with decentralized verification mechanisms without violating the strict latency constraints of smart grid operations. The proposed system provides a scalable, secure, and efficient solution for next-generation energy infrastructures requiring both intelligence and trust.

Future work will focus on deploying the framework on physical edge hardware platforms such as NVIDIA Jetson devices and evaluating real-world energy consumption, while also integrating emerging 6G communication technologies to support ultra-low-latency, large-scale smart grid deployments.

XIV. FUTURE WORK

Future work will focus on deploying the proposed framework on real-world edge computing platforms such as NVIDIA Jetson devices and Raspberry Pi-based IoT gateways to evaluate practical latency, communication overhead, and energy consumption under realistic operating conditions.

A fully distributed blockchain implementation will also be explored to investigate synchronization delays, malicious node resilience, consensus reliability, and scalability across geographically distributed smart grid substations.

In addition, future research will investigate adaptive sequence-learning architectures capable of handling variable-length PMU streams and heterogeneous sampling frequencies without requiring retraining.

Federated learning techniques will also be incorporated to enable privacy-preserving distributed model training across multiple smart grid regions while reducing centralized dependency.

Finally, Explainable Artificial Intelligence (XAI) and self-healing grid control mechanisms will be explored to improve transparency, automated recovery capability, and trustworthiness in next-generation cyber-physical energy infrastructures.

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